ECE448 Spring 2026

L13-pam-theory

UNITED STATES
AIR FORCE
ACADEMY

Nominal Schedule

Course schedule – see the ECE 448 syllabus.

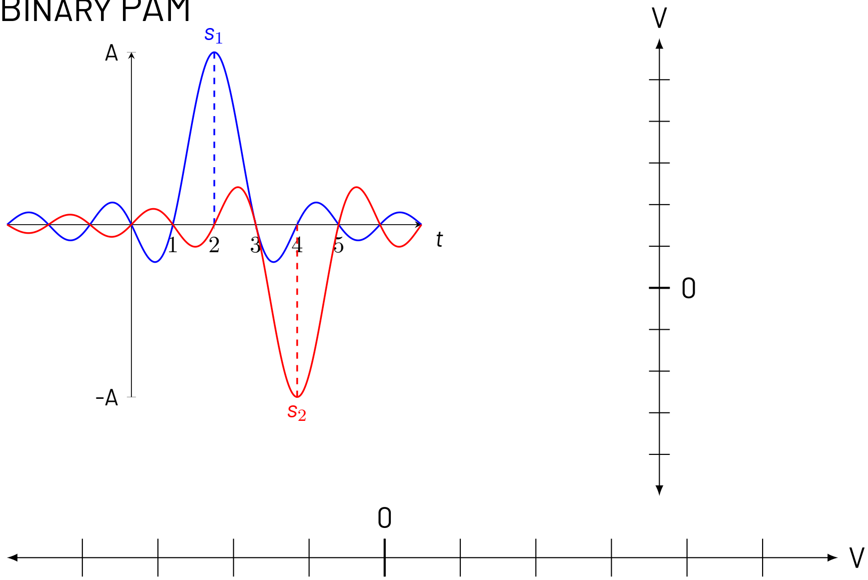
Lesson 3 Learning Objectives

1.1 Understand and articulate the difference between mFSK, mASK, PAM, and mPSK signals.

1.2 Calculate the probability of error for each.

1.3 Draw the constellation diagram for a given modulation technique and

Binary PAM

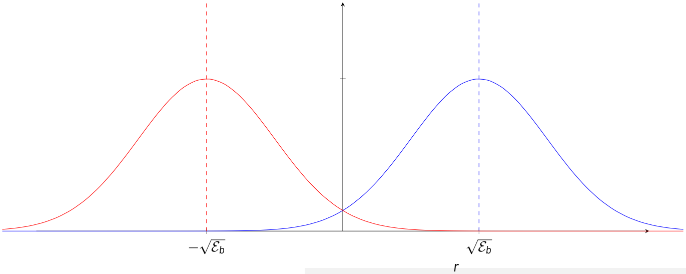


PAM PROBABILITY OF ERROR P_b

- Assume the energy per bit in a PAM pulse is ________________.
- In AWGN channel, we have the received signal as: $r = s_k + n$
- The conditional probability of receiving each pulse:

$$p(r|s_1) = \frac{1}{\sqrt{\pi N_0}} e^{-\left(r-\sqrt{\mathcal{E}_b}\right)^2/N_0}$$

$$p(r|s_2) = \frac{1}{\sqrt{\pi N_0}} e^{-\left(r+\sqrt{\mathcal{E}_b}\right)^2/N_0}$$



PAM P_b

Given a decision threshold of 0, the probability of an error for s_1 tx'd is:

$$P(e|s_1) = \int_{-\infty}^{0} p(r|s_1)dr = \frac{1}{\sqrt{\pi N_0}} \int_{-\infty}^{0} e^{-\left(r-\sqrt{\mathcal{E}_b}\right)^2/N_0} dr$$

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$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} e^{-\left(r-\sqrt{\mathcal{E}_b}\right)^2/N_0} dr = Q\left(\sqrt{\frac{2\mathcal{E}_b}{N_0}}\right)$$

PAM P_b

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Since both signals are equally likely...

PAM Probability of Error

$$P_b = \frac{1}{2}P(e|s_1) + \frac{1}{2}P(e|s_2) = Q\left(\sqrt{\frac{2\mathcal{E}_b}{N_0}}\right)$$

PAM P_b

The energy "distance" between the two signals is $d_{12} = 2\sqrt{\mathcal{E}_b}$, implying:

$$\mathcal{E}_b = \frac{d_{12}^2}{4}$$

PAM P_b

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$$\mathcal{E}_b = \frac{d_{12}^2}{4}$$

Leading to the big result....

PAM Probability of Error

$$P_b = Q\left(\sqrt{\frac{d_{12}^2}{2N_0}}\right)$$

ORTHOGONAL P_b

PAM signals are antipodal, but what if the siganls are orthogonal?

$$s_1 = \left(\sqrt{\mathcal{E}_b}, 0\right)$$
$$s_2 = \left(0, \sqrt{\mathcal{E}_b}\right)$$

Orthogonal Signals P_b

$$P_b = Q\left(\sqrt{\frac{\mathcal{E}_b}{N_0}}\right)$$

$m-$ary PAM P_b

$m-$ary PAM signals

$$s_m = \sqrt{\mathcal{E}_g} A_m$$

$m-$ary PAM

$$P_m = \frac{2\left(M-1\right)}{M} Q\left(\sqrt{\frac{6 \ln\left(M\right) \mathcal{E}_{b,avg}}{\left(M^2-1\right) N_0}}\right)$$

$m-$ARY PAM P_b (BAND-LIMITED)

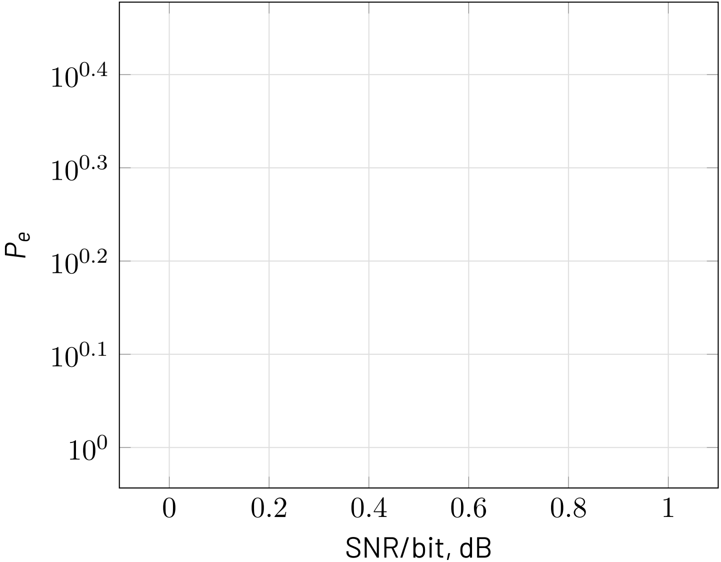
$m-$ary PAM signals

$$s_m = \sqrt{\mathcal{E}_g} A_m$$

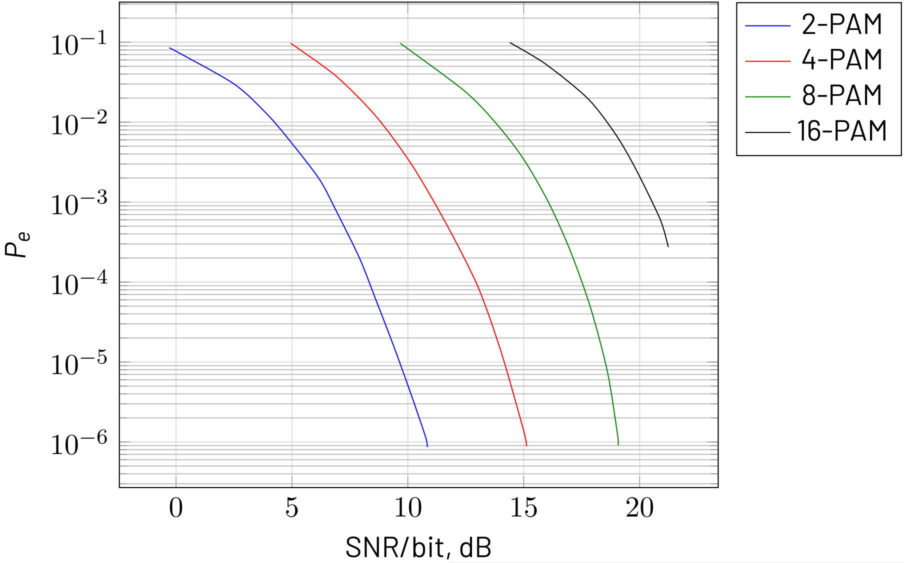
$m-$ary PAM

$$P_m = \frac{2\left(M-1\right)}{M} Q\left(\sqrt{\frac{6\mathcal{E}_{b,\text{avg}}}{\left(M^2-1\right)N_0}}\right)$$

Binary Signals

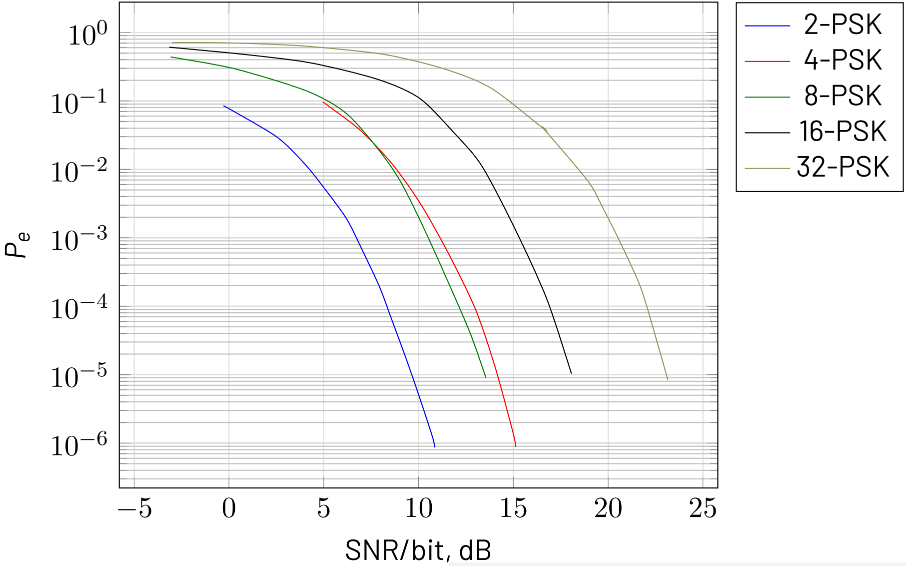


PAM Signals



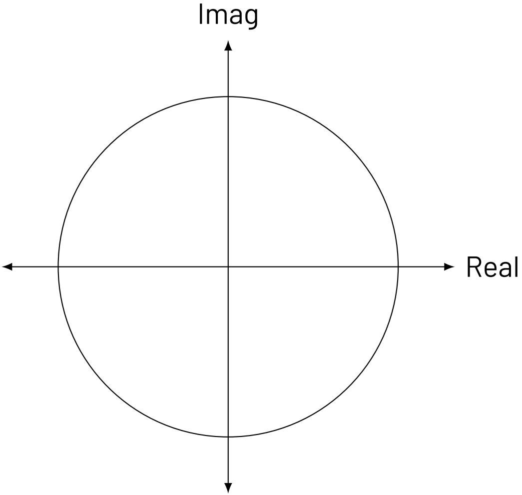
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
P_e
0
5
10
15
20
SNR/bit, dB
2-PAM
4-PAM
8-PAM
16-PAM

PSK SIGNALS

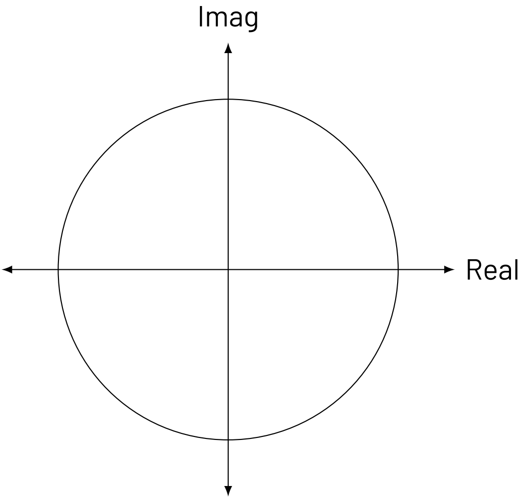


10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
P_e
-5
0
5
10
15
20
25
SNR/bit, dB
2-PSK
4-PSK
8-PSK
16-PSK
32-PSK

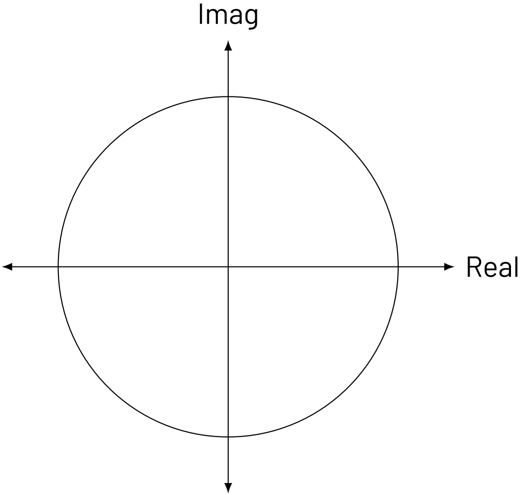
Constellation Diagram: 2-PAM



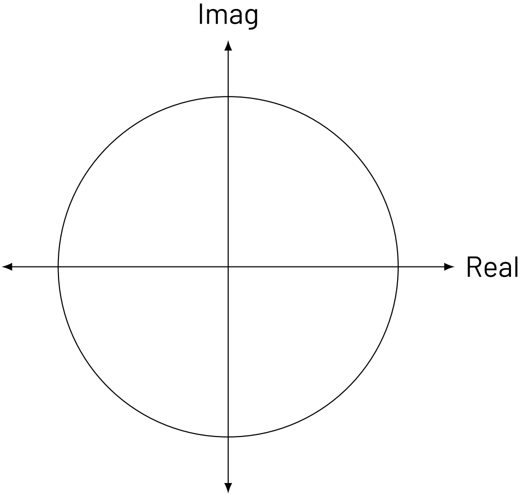
Constellation Diagram: 4 PAM



Constellation Diagram: m-PSK



Gray Coding



Gray Coding

Each successive symbol varies only by 1 bit.

Map out Gray Code for 8-PSK